Operating Systems
Lab 8

1. Write a C program as follows. The program receives a number from command line when a user runs the program. The parent then forks a child to calculate the summation from 1 to the number that the user specified. The child sends the result back to the parent via the pipe. The parents then prints the result.

Note: One way to convert an integer to string is to use sprintf() function as follows

sprintf(str, “%d”, sum);

This statement converts the integer store in the variable sum to string and store the result in the variable str.

The example output of the program is as follows:
./lab8_pipe 5
Parent: waiting for my child
Child: I am calculating
Child: the result is 15
Child: I am sending data
Child: Goodbye
Parent: sum from my child is 15
Parent: Goodbye

2. Write a C program to do the same task as question number 1 but use share memory. The parent must remove the shared memory before existing.

The example output of the program is as follows:
./lab7_shm 5
Parent: I have created a shared memory for result...
Parent: I have attached the shared memory...
Parent: I am about to fork a child process...
Parent: Waiting for my child
Child: I am calculating
Child: The result is 15
Child: Goodbye
Parent: sum from my child is 15
Parent: I have detached the shared memory...
Parent: I have removed the shared memory...
Parent: Goodbye